1. $y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + \frac{1}{4}x(n) + \frac{1}{2}x(n-1)$

(a). Using z transform,

$$Y(z) = \frac{5}{6}z^{-1}Y(z) - \frac{1}{6}z^{-2}Y(z) + \frac{1}{4}X(z) + \frac{1}{2}z^{-1}X(z)$$

$$Y(z)\left(1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}\right) = X(z)\left(\frac{1}{4} + \frac{1}{2}z^{-1}\right)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{4} + \frac{1}{2}z^{-1}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}$$

(b) $H(\Omega) = H(z)\big|_{z=e^{j\Omega}}$

$$\Rightarrow H(\Omega) = \frac{\frac{1}{4} + \frac{1}{2}e^{-j\Omega}}{1 - \frac{5}{6}e^{-j\Omega} + \frac{1}{6}e^{-j2\Omega}}$$

The amplitude and phase response can be seen from the figures.

(c) $$H(z) = \frac{\frac{1}{4} + \frac{1}{2}z^{-1}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}} = \frac{\frac{1}{4}z^2 + \frac{1}{2}z}{z^2 - \frac{5}{6}z + \frac{1}{6}}$$

$$\frac{H(z)}{z} = \frac{\frac{1}{4}z + \frac{1}{2}}{z^2 - \frac{5}{6}z + \frac{1}{6}} = \frac{\frac{1}{4}z + \frac{1}{2}}{(z - \frac{1}{3})(z - \frac{1}{2})} = \frac{A}{z - \frac{1}{3}} + \frac{B}{z - \frac{1}{2}}$$

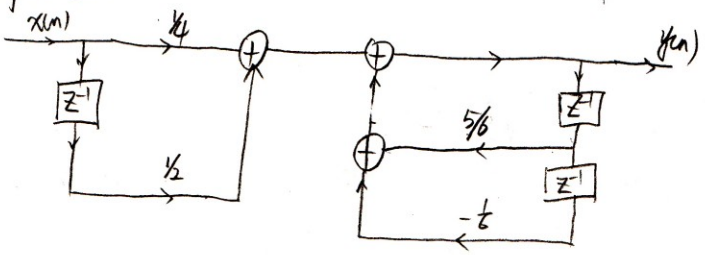
$$A = (z - \tfrac{1}{3})\frac{H(z)}{z}\Big|_{z=\frac{1}{3}} = \frac{\frac{1}{4}z + \frac{1}{2}}{(z - \frac{1}{2})}\Big|_{z=\frac{1}{3}} = \frac{\frac{1}{4}\cdot\frac{1}{3} + \frac{1}{2}}{\frac{1}{3} - \frac{1}{2}} = -\frac{7}{2}$$

$$B = (z - \tfrac{1}{2})\frac{H(z)}{z}\Big|_{z=\frac{1}{2}} = \frac{\frac{1}{4}z + \frac{1}{2}}{(z - \frac{1}{3})}\Big|_{z=\frac{1}{2}} = \frac{\frac{1}{4}\times\frac{1}{2} + \frac{1}{2}}{\frac{1}{2} - \frac{1}{3}} = \frac{15}{4}$$

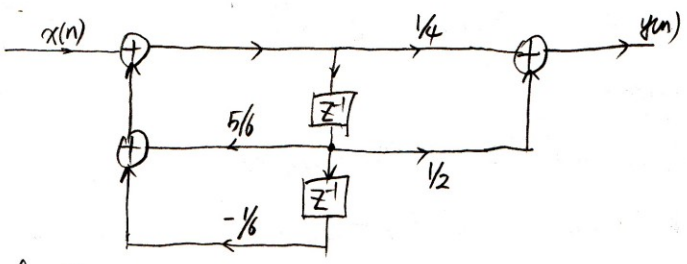
$$\Rightarrow H(z) = -\frac{7}{2}\cdot\frac{z}{(z - \frac{1}{3})} + \frac{15}{4}\cdot\frac{z}{z - \frac{1}{2}}$$

$$h(n) = -\frac{7}{2}\left(\frac{1}{3}\right)^n u(n) + \frac{15}{4}\cdot\left(\frac{1}{2}\right)^n u(n)$$

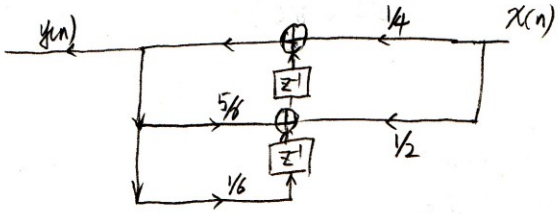
(d). direct-form I.



Direct-form II.



transformed direct-form II.



(e). Both poles are inside the unit circle, so the system is stable.

2. $h(n) = [2, -1, 3, -1, 2]$

(a) $H(z) = \sum_{n=0}^{\infty} h(n) z^{-n} = 2 - z^{-1} + 3z^{-2} - z^{-3} + 2z^{-4}$

(b) $H(\omega) = H(z)|_{z=e^{j\omega}} = 2 - e^{-j\omega} + 3e^{-j2\omega} - e^{-j3\omega} + 2e^{-j4\omega}$

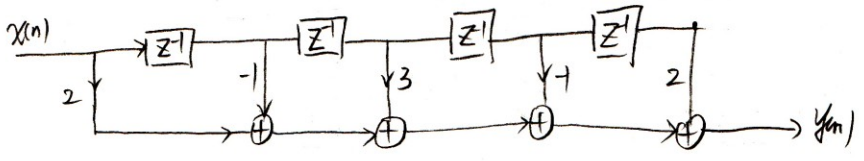
· amplitude and phase response.

(c) $H(z) = \frac{Y(z)}{X(z)} = 2 - z^{-1} + 3z^{-2} - z^{-3} + 2z^{-4}$

$Y(z) = 2X(z) - z^{-1}X(z) + 3z^{-2}X(z) - z^{-3}X(z) + 2z^{-4}X(z)$

$\Rightarrow y(n) = 2x(n) - x(n-1) + 3x(n-2) + x(n-3) + 2x(n-4)$

(d). Direct form-I.



(e) Because, $\sum_{n=-\infty}^{\infty} |h(n)|^2 < \infty$, the system is stable.

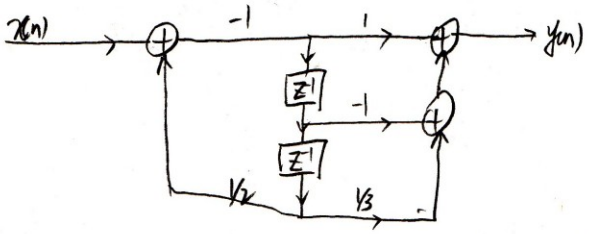
3. $H(z) = \dfrac{1 - z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{1}{2}z^{-2}}$

(a) $H(z) = \dfrac{Y(z)}{X(z)} = \dfrac{1 - z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{1}{2}z^{-2}}$

$\Rightarrow Y(z) - \frac{1}{2}z^{-2}Y(z) = X(z) - z^{-1}X(z) + \frac{1}{3}z^{-2}X(z)$

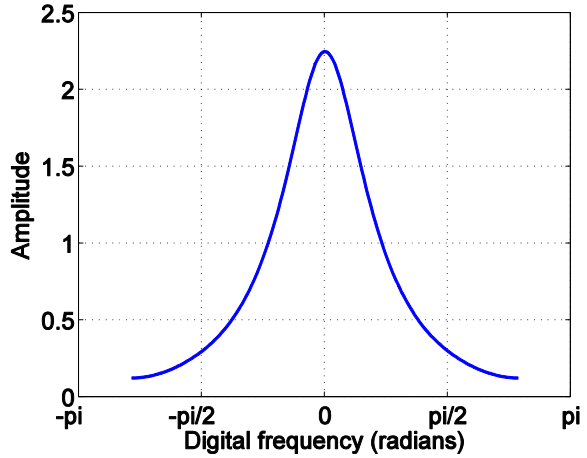
$y(n) - \frac{1}{2}y(n-2) = x(n) - x(n-1) + \frac{1}{3}x(n-2)$

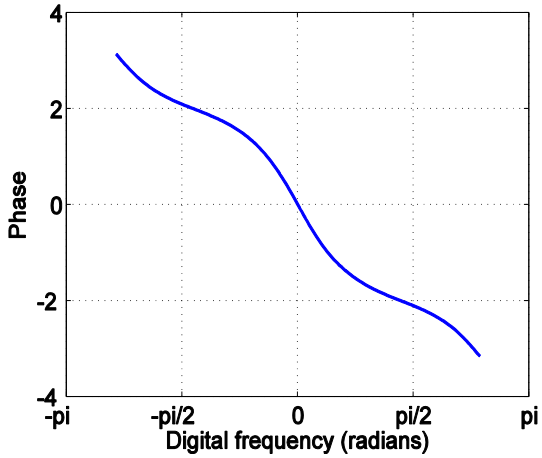
(b) Direct form - II.



(c). Both poles are inside the unit circle, so the system is stable.

1. (b) amplitude and phase response.





2 .(b) amplitude and phase response.

